

HapticCompass: Body-Movement Guidance Using Hanger Reflex and Grain-based Vibrotactile Rendering

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Abstract. We present *HapticCompass*, a head-worn haptic system that guides user head orientation through intuitive, body-driven cues. In this demo, participants experience how subtle haptic signals guide head movement and support orientation without visual guidance. The system combines two mechanisms: a hanger reflex that triggers head rotation, and vibrotactile feedback that supports continuous adjustment. In the demo, two roles are involved. The *follower* follows haptic cues to complete rotation tasks based on bodily perception, while the *controller* specifies target orientations through a custom graphical user interface (GUI). This interactive setup highlights how haptic guidance supports fast, precise, and low-effort orientation guidance without requiring explicit visual instruction.

Keywords: Interpersonal Embodied Interaction · Head Rotation Guidance · Hanger Reflex · Grain-based Vibrotactile Rendering.

1 Demo Overview

Movement guidance is often achieved through direct physical intervention or explicit visual instruction. While effective, these approaches either impose strong control over the body or require active interpretation, making them less suitable for lightweight and intuitive interaction. This limitation becomes particularly pronounced in remote contexts, such as remote physiotherapy, where practitioners cannot provide the hands-on guidance that is often necessary [1]. This demo explores an alternative form of guidance based on bodily perception, where both directional intent and target angles are conveyed through haptic cues.

To make this interaction tangible, the demo is structured as a two-person experience. One person (controller) specifies target directions and angles through a custom GUI, while the other person (follower), without any visual information, relies entirely on haptic feedback to interpret these commands and adjust their head orientation. By varying the feedback conditions, the demo enables exploration of how different forms of haptic guidance influence movement, ranging from subtle suggestions to more compelling directional cues.

2 System & Demo Setup

The demo consists of a head-mounted haptic interface worn by the follower and a GUI used by the controller. The follower only receives haptic guidance information during the interaction. As shown in Fig. 1, the head-mounted interface integrates two types of actuators. Four pneumatic pouches placed in the head-band apply pressure to the front and back sides of the head to induce rotation via the hanger reflex. Two vibration actuators positioned above the ears provide grain-based vibrotactile feedback, where vibration frequency and amplitude vary with the angular distance to the target.

The controller operates a GUI on a laptop and selects feedback modes (hanger reflex, vibration, or combined) and specifies target angles within a range of -60° to 60° . The haptic interface delivers corresponding real-time guidance based on these inputs. Each session involves two participants and lasts approximately 3–5 minutes. Participants may switch roles after each session to experience both roles.

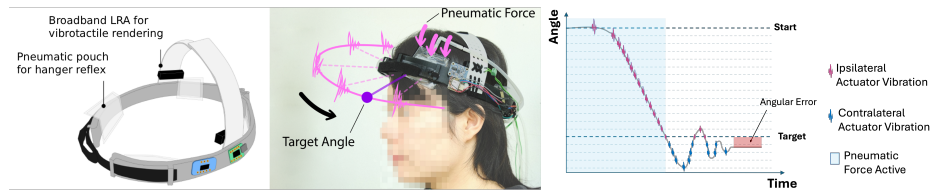


Fig. 1. Overview of the HapticCompass system. Left: head-mounted interface with pneumatic actuators for the hanger reflex and LRA actuators for vibrotactile feedback. Middle: a user wearing the device and being guided toward a target head orientation. Right: example head rotation trajectory showing how haptic feedback supports alignment with the target angle over time.

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